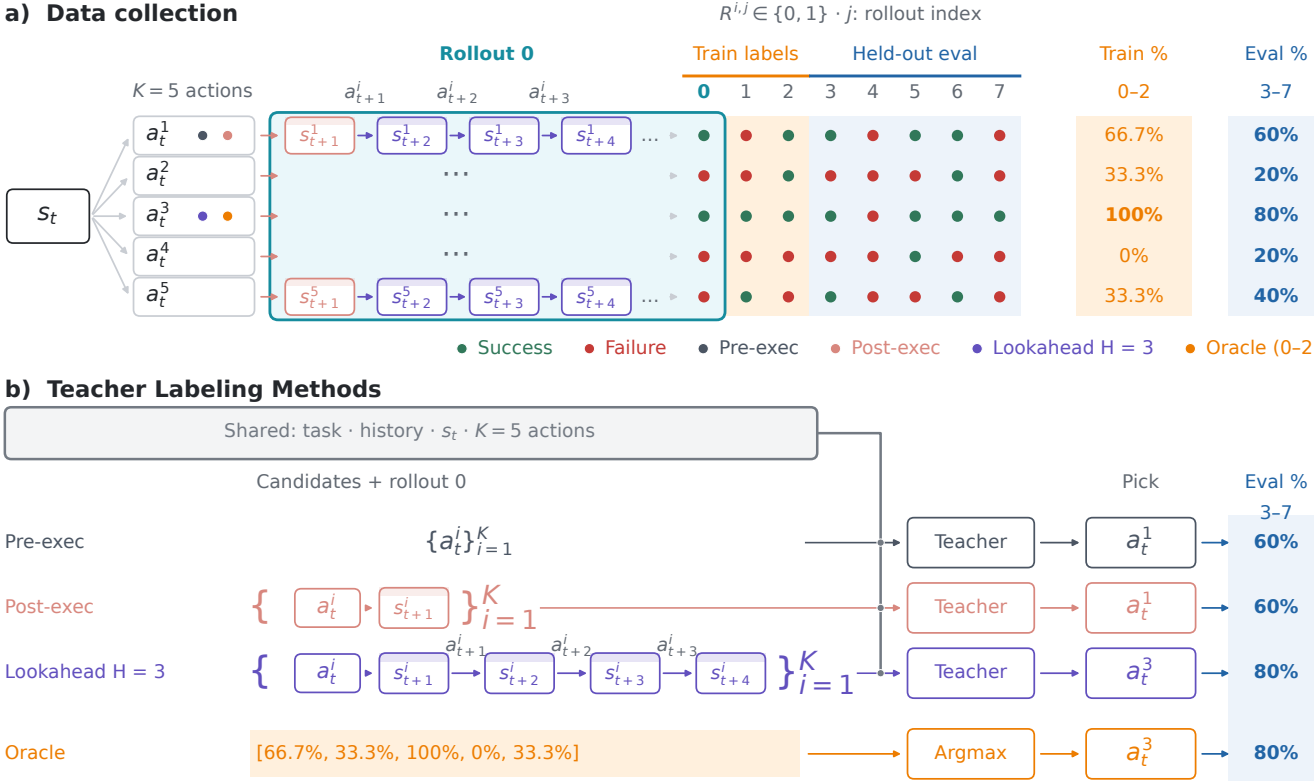
a) Data collection
$R^{i,j} \in \{0,1\}$ · j: rollout index
Rollout 0
Train labels
Held-out eval
Train %
Eval %
$K = 5$ actions
0 1 2 3 4 5 6 7
0–2
3–7
s_t
66.7%
60%
33.3%
20%
100%
80%
0%
20%
33.3%
40%
Success
Failure
Pre-exec
Post-exec
Lookahead H = 3
Oracle (0–2)
b) Teacher Labeling Methods
Shared: task · history · s_t · $K = 5$ actions
Candidates + rollout 0
Pick
Eval %
3–7
Pre-exec
Post-exec
Lookahead H = 3
Oracle
[66.7%, 33.3%, 100%, 0%, 33.3%]
Teacher
Teacher
Teacher
Argmax
60%
60%
80%
80%